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Comprehensive blood glucose level prediction from HbA_{1c} levels using machine learning models across the biological range

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Accurate blood glucose prediction from HbA_{1c} is crucial for personalized health monitoring and diabetes management. Existing models often overlook data imbalances and hyperglycemic outliers, reducing accuracy. This study proposes a machine-learning framework to address these challenges. This study investigates a dataset of 197,180 patient samples, focusing on key features such as age, glucose, and HbA_{1c} levels. The performance of 42 machine learning models was evaluated using Kernel Density Estimates (KDE) analysis and targeted oversampling to enhance model robustness. A Flask web application has been developed for deployment on the Heroku platform. Results show that the MLP Regressor achieved a moderate R^2 with the raw data. Logarithmic transformation effectively reduced RMSE. Gaussian KDE identified a low-density region around 550 mg/dL in hyperglycemia, prompting targeted oversampling. This, combined with logarithmic transformation, resulted in an R^2 of 0.93 and the lowest RMSE with the LGBM model, indicating strong predictive robustness for blood glucose levels. The proposed approach effectively enhances glucose level prediction accuracy in both healthy and diabetic management. The Heroku-deployed web app provides an accessible tool for clinicians and individuals, supporting real-time diabetes management and personalized health monitoring.

Keywords Glucose prediction, HbA_{1c}, Machine learning, Data augmentation, Transformation

Blood glucose levels are a fundamental parameter in the management and diagnosis of diabetes, acting as a direct measure of an individual's metabolic health[1]. Regular follow-up of blood glucose is vital for the prevention of both acute complications such as hypoglycemia, and chronic conditions like neuropathy, retinopathy, and cardiovascular diseases, all of which are associated with diabetes[2]. Predicting actual glucose levels and fluctuations can significantly enhance clinical outcomes by facilitating timely interventions and enabling personalized treatment strategies tailored to individual patient needs[3]. Accurate glucose prediction not only supports the optimization of diabetes treatment but also informs decisions related to dietary adjustments and lifestyle changes[4].

Predicting glucose levels using glycated hemoglobin (HbA_{1c}), a reflection of average blood glucose for the last two to three months, is widely used in clinical practice to gauge long-term glucose control[5]. However, as pointed out by Marling et al., HbA_{1c} alone does not provide real-time insights into daily glucose fluctuations, which are crucial for timely interventions and personalized treatment plans[6]. Therefore, leveraging novel models that integrate HbA_{1c} with real-time glucose measurements utilizing additional demographic or lifestyle factors offers a more comprehensive approach to predicting glucose variability. Linear regression is one of the most widely used statistical models for predicting estimated average blood glucose (eAG) levels from HbA_{1c} due to the relatively strong and linear relationship between these variables. R-squared (R^2) score was found 0.84 using Linear regression for a dataset including only 2,700 patient samples. However, further validation and exploration of more complex models could be considered as the dataset size becomes larger[7].

Recent advancements in machine learning (ML) techniques have provided researchers with powerful tools to analyze complex and multidimensional datasets, uncovering hidden patterns that traditional statistical methods often overlook[8]. In the context of diabetes management, several studies have demonstrated the utility of ML in predicting blood glucose levels based on a variety of clinical and demographic parameters specifically for Type 1

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and 2 Diabetes[9–12]. These approaches can transform HbA_{1c} from a static measure into a more dynamic predictive tool, enabling people to create a personalized health model by anticipating glucose fluctuations.

Hypoglycemia and hyperglycemia are critical conditions characterized by blood sugar levels falling below 70 mg/dL and rising above 130 mg/dL, respectively. These events are less frequent than normal glucose levels, resulting in a limited number of samples that can hinder the model's ability to learn effectively from extreme cases of blood glucose levels[13–15]. Data augmentation, a technique widely used in fields like medical image segmentation and disorder classification, has also demonstrated its potential to significantly improve the performance of ML models in imbalanced biological data[16].

This paper investigates the improvement of glucose prediction from HbA_{1c} levels using ML algorithms by investigating data transformation and the effect of oversampling by repeating. Unlike prior studies that relied on smaller, often imbalanced datasets, our work leverages a significantly larger dataset and introduces novel preprocessing techniques like KDE-informed oversampling, logarithmic transformation, and targeted augmentation. This approach effectively mitigates the limitations associated with extreme cases of hypo- and hyperglycemia, enabling higher model robustness and predictive accuracy. Through this investigation, we aim to highlight the importance of ML models in the field of health management and inform future guidelines on the relevance of eAG reporting. Figure 1 graphically represents the workflow of the study.

Results

EDA

The Glucose vs. HbA_{1c} plot exhibits a right-skewed distribution, with most of the points clustered in the lower-left region of the plot. The data points are represented in varying shades of the same color, corresponding to different age groups (25–100 years), providing insights into the relationship between age and key metabolic markers: glucose and HbA_{1c} levels. The observed distribution indicates that older individuals (aged 75–100) tend to exhibit higher glucose and HbA_{1c} values compared to younger individuals (aged 25–50) (Fig. 2A).

While glucose shows a robust correlation with HbA_{1c}, its relationships with protein and microalbumin levels are notably weaker. The correlation between glucose and insulin ($r = 0.11$) points to another weak positive correlation. Consequently, these weak correlations were not included in the subsequent ML analysis (Fig. 2B).

Machine learning on raw data

The analysis included multiple preprocessing steps and transformations to optimize model performance for glucose prediction. The initial dataset was divided into four chunks, and 42 ML models were trained and evaluated on each chunk, with performance assessed on the raw data (Table 1). The use of LazyPredict allowed for an efficient and comprehensive comparison of various regression models, including Multilayer Perceptron (MLP) Regressor, ElasticNetCV, Lasso, and potentially others. This approach helped identify the MLP Regressor as the optimal model for predicting the target variable glucose levels across the different data chunks of raw data.

The R^2 values range from 0.68 to 0.71, indicating that the model explains a moderate portion of the variance in the target variable. The RMSE values, ranging from 21.13 to 22.36, reveal that, on average, the model's predictions deviate by about 21–22 glucose units. Then it is aimed to explore fine-tuning the model's hyperparameters to further reduce RMSE and optimize its performance by achieving a higher R^2 score.

Data transformation

Different transformation algorithms were applied to the glucose values to improve model performance as summarized in Table 4. Figure 3 demonstrates the distribution effect of different transformation algorithms applied to the glucose variable of the data. The raw data has the histogram of a heavy right skew, with a long tail extending toward higher glucose levels. This type of imbalance in data can significantly hinder model performance, necessitating data transformation[17]. The data is biased towards majority class normglycemia. Among the various

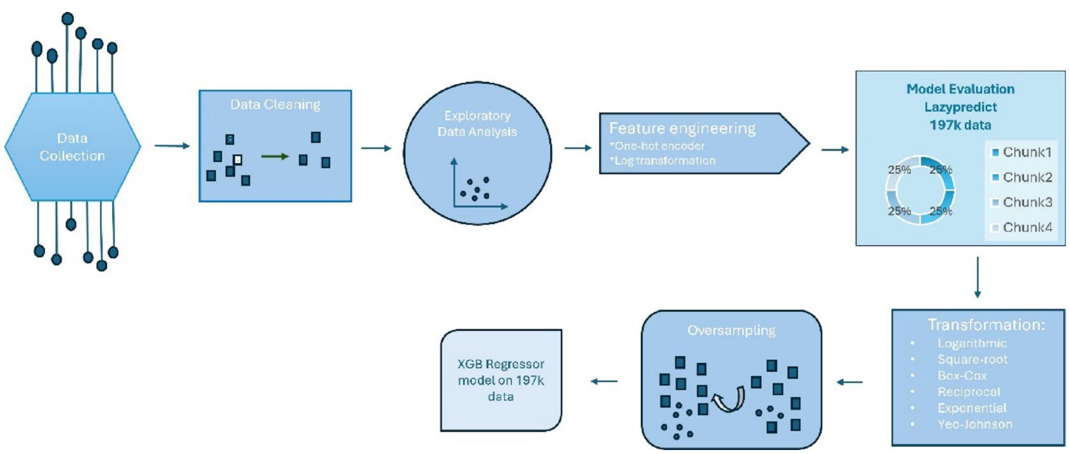


Fig. 1. A work flowchart of this study.

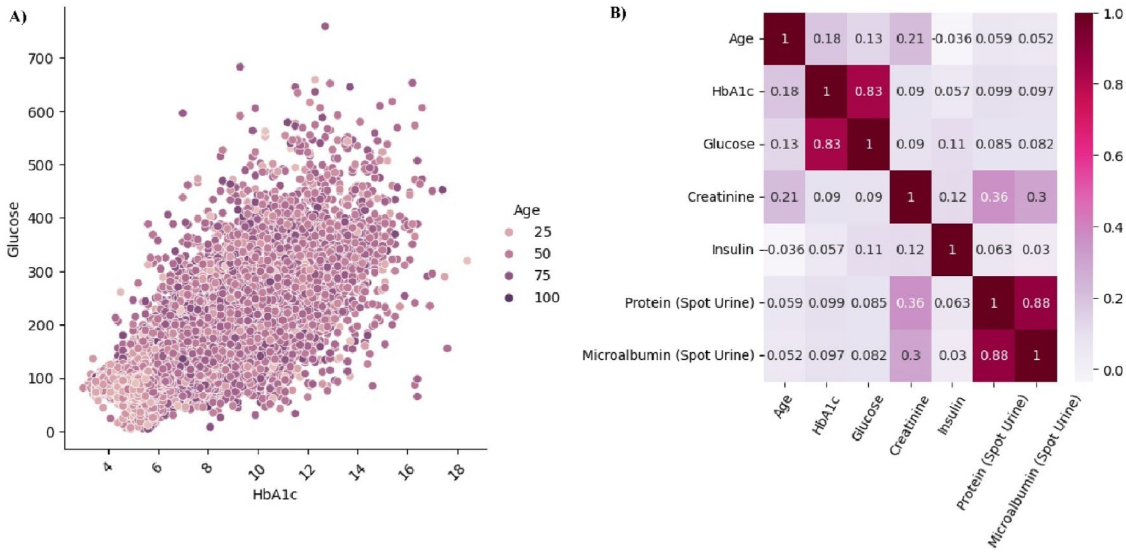


Fig. 2. (**A**) Relationship between HbA_{1c} and glucose levels by age group (**B**) Correlation matrix of metabolic markers and demographic factor age.

Chunk	Model	R^2	RMSE	Time taken
Chunk 1	MLPRegressor	0.68	21.59	4.84
Chunk 2	MLPRegressor	0.71	21.29	4.15
Chunk 3	MLPRegressor	0.69	22.26	3.55
Chunk 4	MLPRegressor	0.69	21.13	4.56

Table 1. Performance comparison of regression models on glucose prediction using raw data.

transformations applied to the glucose values, both logarithmic and square root transformations appeared to yield a more normally distributed dataset.

Model performance of transformed data

Table 2 shows performance metrics for various ML models applied to the transformed glucose data of 197 k records. Logarithmic transformed based on R^2 (0.68) demonstrates the best predictive performance with lower RMSE using Gradient Boosting Regressor.

Square transformed with MLP Regressor also perform well (R^2: 0.68) but has a high RMSE value. Exponential transformed appears unsuitable for this dataset failing to produce valid results (indicated by "NaN" and "inf" values).

Kernel density estimation on glucose data

KDE function calculated the density of data points numerically, a prominent peak in the high glucose region (around 500 mg/dL), indicating a high density of observations in this range (Fig. 4). Additionally, the KDE identifies glucose values above 550 mg/dL as part of a low-density region or underrepresented, highlighting areas with fewer data points.

The KDE plot illustrates the distribution of glucose levels across different categories: hypoglycemia, normoglycemia, and hyperglycemia. The density curves indicate that most glucose values are concentrated in the lower range (< 130 mg/dL), highlighting a significant prevalence of normoglycemic readings.

Oversampling impact on descriptive statistics

Descriptive statistics were used to summarize the dataset. The mean age of the participants was calculated as 46.21 years, with a standard deviation of 17.73 years. The gender distribution was assessed by calculating the female-to-male ratio, which was 2.49, indicating a predominance of female participants in the study cohort (Supplementary Figure S1).

KDE plots were generated to visualize density differences in glucose and HbA_{1c} values after oversampling. The box plot on Fig. 5A and B displays the distribution of the original glucose values, with the mean, median, and key percentiles (Q1, Q2, Q3) indicated. The box plot on the right shows the distribution of the glucose values, generated through oversampling technique. Compared to the original data, the oversampled dataset appears to have a higher Q3 value indicating that a larger portion of the data falls within a higher glucose range.

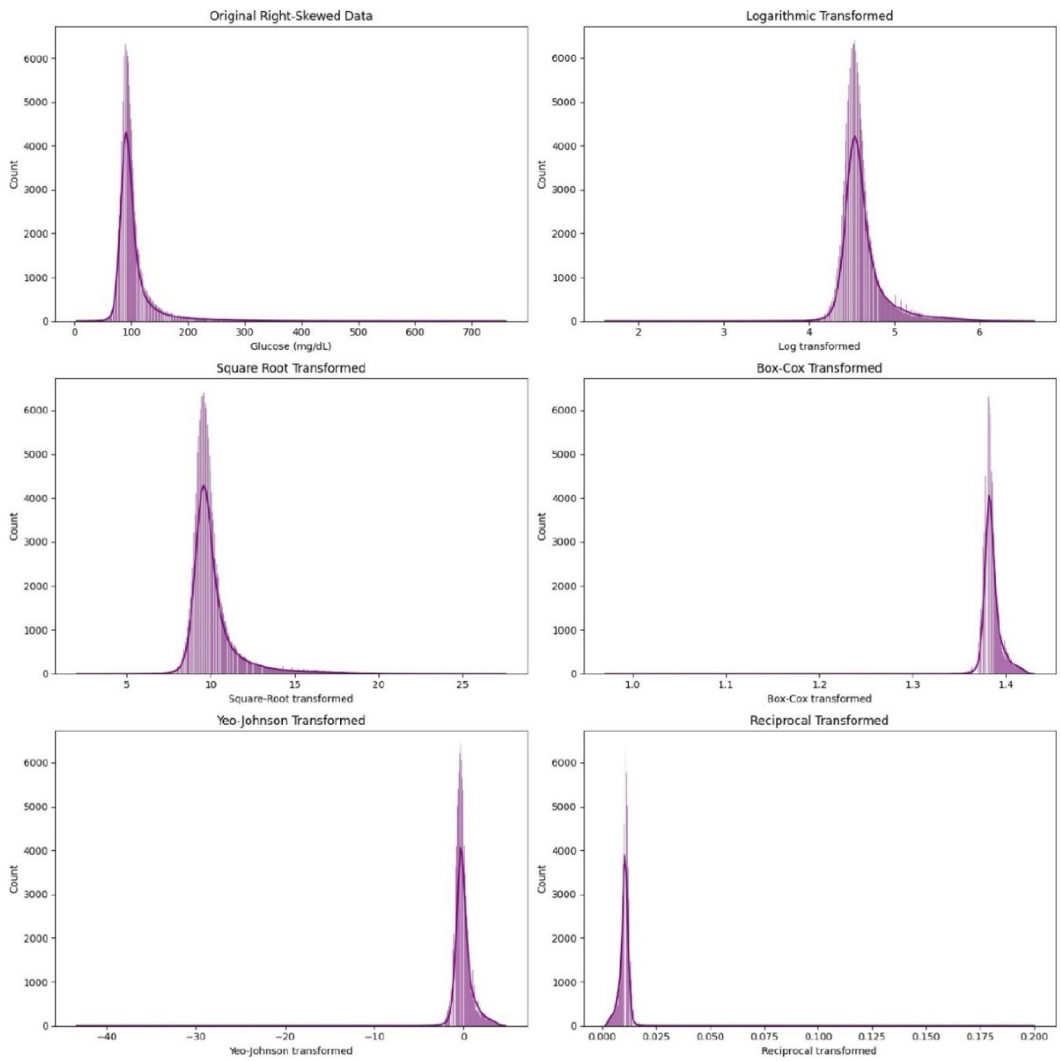


Fig. 3. Comparison of different data transformation algorithms on the glucose distributions.

Transformation	Model	R^2	RMSE	Time taken
Logarithmic transformed	GradientBoostingRegressor	0.65	0.15	0.95
Square Root transformed	MLPRegressor	0.68	0.86	3.07
Box-cox transformed	GradientBoostingRegressor	0.57	0.01	0.96
Yeo-Johnson transformed	GradientBoostingRegressor	0.57	0.64	1

Table 2. Performance comparison of machine learning models on glucose prediction transformed data(197 K) using different algorithms.

The distributional changes in glucose (mg/mL) and HbA_{1c} (%) before and after oversampling are illustrated in Supplementary Figure S2, which compares the original (A, B) and oversampled (C, D) datasets.

The mean glucose value in the original dataset is 146.91 with a standard deviation of 17.68, showing some variability in glucose levels. The augmented dataset has a slightly lower mean of 146.97, suggesting that the augmentation did not significantly alter the average glucose level.

Model performance on oversampled data

After oversampling the data, we performed 42 different ML models on glucose prediction for 4 chunks. Overall, oversampling on logarithmic transformed glucose data yielded all models demonstrated strong predictive capabilities with high R^2 values and low RMSEs. The Light Gradient Boosting Machine (LGBM) Regressor appears to be the most accurate in the final chunk with the lowest RMSE (Table 3).

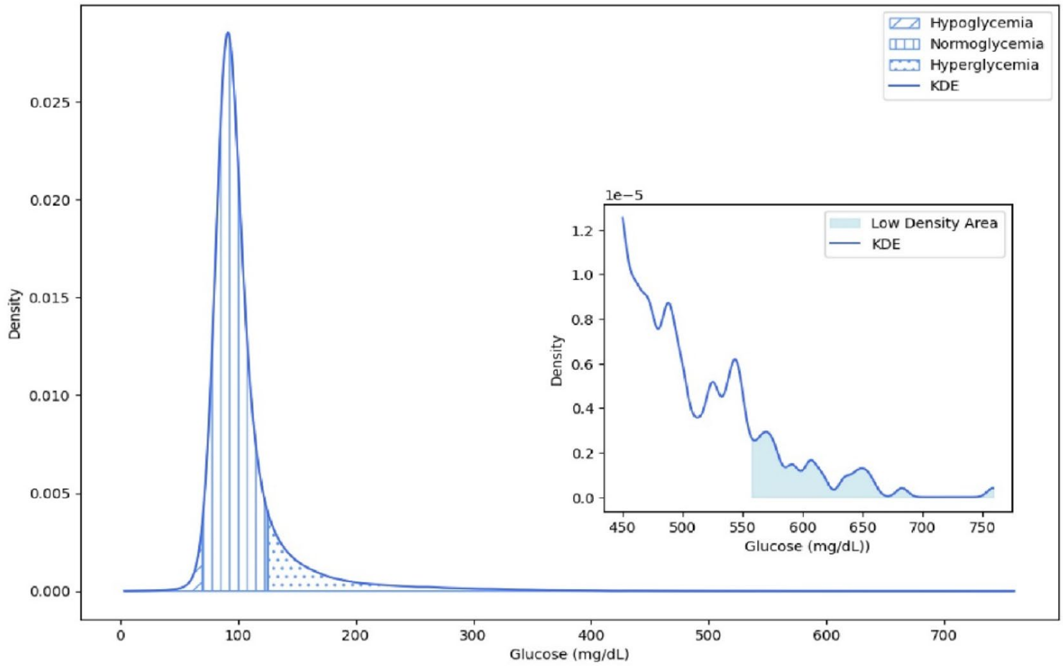


Fig. 4. Kernel Density Estimation of glucose values determines low density regions, inset graph shows the zoom-in (450–750).

To ensure fair comparison, we included Linear Regression as a traditional statistical baseline, against which more complex models like Gradient Boosting and LGBM were evaluated. Figure 6 includes a comparison of Linear Regression on both the original dataset (A) and the oversampled data (E), the latter consisting of 215,000 patient records. While performance improves modestly with oversampling (R^2: 0.80 vs. 0.69), the RMSE remains considerably higher compared to the more advanced LGBM model (D), demonstrating the limitations of linear models in capturing the non-linear relationships found in glucose prediction. Additionally, for exploratory purposes, we trained models to predict HbA_{1c} from glucose levels. The results of this reverse prediction are presented in Supplementary Figure S3.

Additionally, fivefold cross-validation yielded RMSE values of 0.15, 0.15, 0.15, 0.15, and 0.50 for each fold, with an average RMSE of 0.22. While the model demonstrated strong performance across most folds, the significantly higher RMSE in the fifth fold suggests potential issues with specific data points. Supplementary Figure S4 presents the learning curves (RMSE) for the LGBM Regressor trained on the dataset using fivefold cross-validation. The figure illustrates the model's performance on both the training and validation sets across increasing training sizes. The learning curves indicate that the model achieves stable and converging performance as the training set size increases. The training RMSE remains low (~0.14) and validation RMSE converges (~0.22–0.23), with a small gap between the two curves, suggesting the model is not overfitting. The narrow confidence bands further indicate low variance across folds, supporting the model's generalization capability.

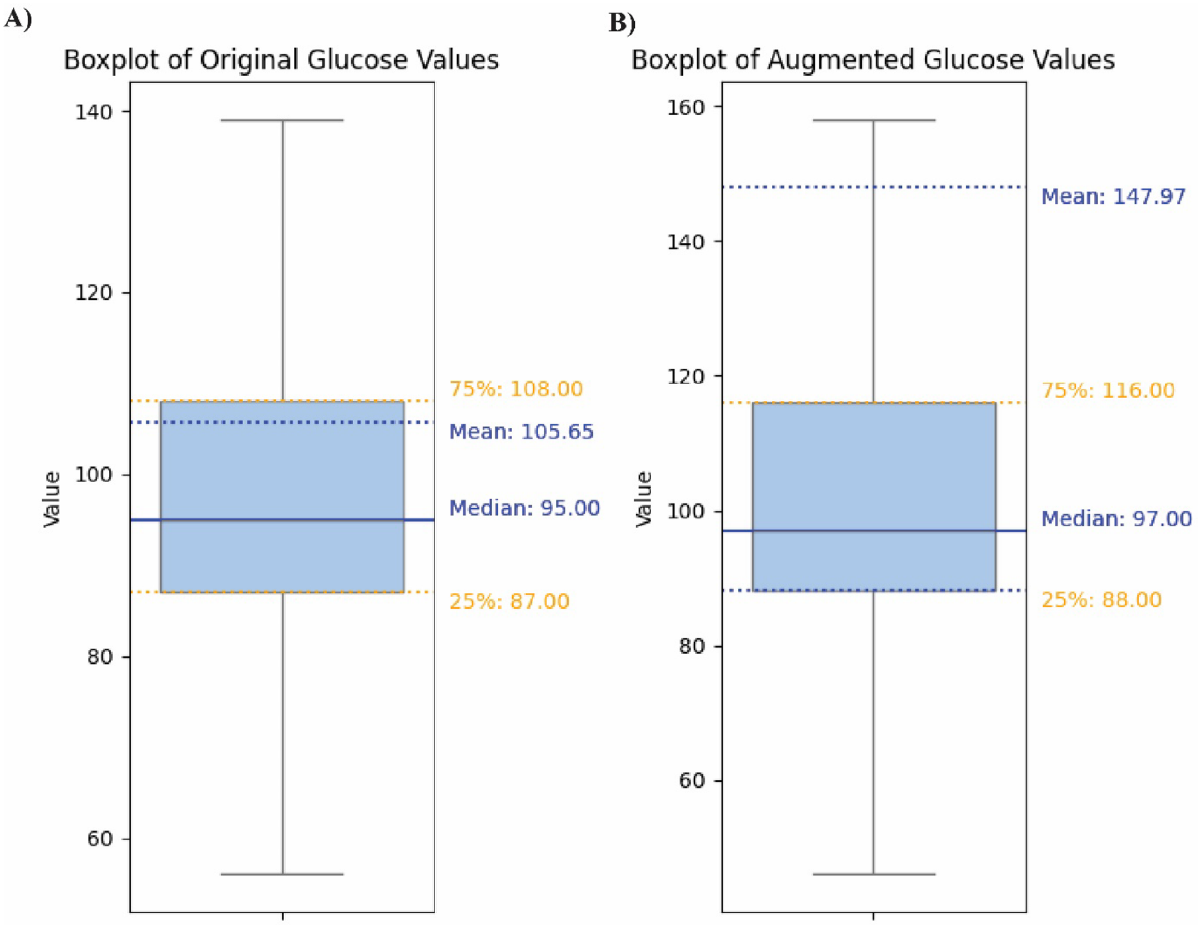
Implementation of the model

The Glucose Prediction App, designed to estimate blood glucose levels based on HbA_{1c} measurements, has been successfully implemented as a web application which is accessible at: https://glucosehba1c-02dd36d2b397.herokuapp.com/.

Discussion

Blood glucose levels are crucial indicators of metabolic health and are directly linked to conditions such as diabetes, obesity, and cardiovascular diseases according to World Health Organization (WHO)[18]. Accurate prediction of blood glucose levels across the entire biological range is vital for diagnosis and treatment of these diseases. To the best of our knowledge, no study has been conducted on modelling blood glucose levels from HbA_{1c} levels for the entire biological range. Several studies have focused on predicting blood glucose levels specifically for Type I or Type II diabetes mellitus patients, often limiting their scope to ML classification tasks. Our study demonstrates the effectiveness of tailored data transformations, balanced data augmentation, and advanced ML models in predicting blood glucose levels with high accuracy.

From the feature screening results, the positive correlation between glucose and HbA_{1c} is well-established with correlation 0.83. Similarly, Del Parigi A et al. showed that HbA_{1c} was as a significant predictor using ML algorithms in patients with diabetes[19]. The correlation matrix indicates that age and insulin have a weak relationship with glucose levels, while there is not a direct relationship between glucose and protein excretion in urine. Since the data encompasses the full biological range, it likely includes individuals across the spectrum



C)

Original dataset	count	mean	std	min	25%	50%	75%	max
Age	197180	46.21	17.73	1	32	47	60	124
Glucose	197180	105.65	38.93	4	87	95	108	759
HbA1c	197180	5.92	1.18	3	5.3	5.6	6	18.4
Oversampled dataset	**count**	**mean**	**std**	**min**	**25%**	**50%**	**75%**	**max**
Age	215680	46.93	17.85	1	33	48	60	124
Glucose	215680	147.97	143.68	4	88	97	116	759
HbA1c	215680	6.48	2.24	3	5.3	5.7	6.3	18.4

Fig. 5. Comparison of original and oversampled glucose data statistics.

Chunk	Model	R^2	RMSE	Time taken
1	XGBRegressor	0.93	0.15	1.26
2	HistGradientBoostingRegressor	0.94	0.15	4.19
3	LGBMRegressor	0.94	0.15	46.34
4	LGBMRegressor	0.94	0.14	18.14

Table 3. Performance comparison of regression models on glucose prediction using transformed augmented big data.

from normoglycemia to overt diabetes and may mask the characteristic strong positive correlation between glucose and insulin that is typically seen in Type 2 Diabetes population[20].

The trend aligns with established findings in the literature that aging is associated with an increase in fasting glucose levels, partly due to changes in pancreatic β-cell function and glucose metabolism efficiency. For instance, the study suggests that with advancing age, blood glucose levels generally increased with Lasso

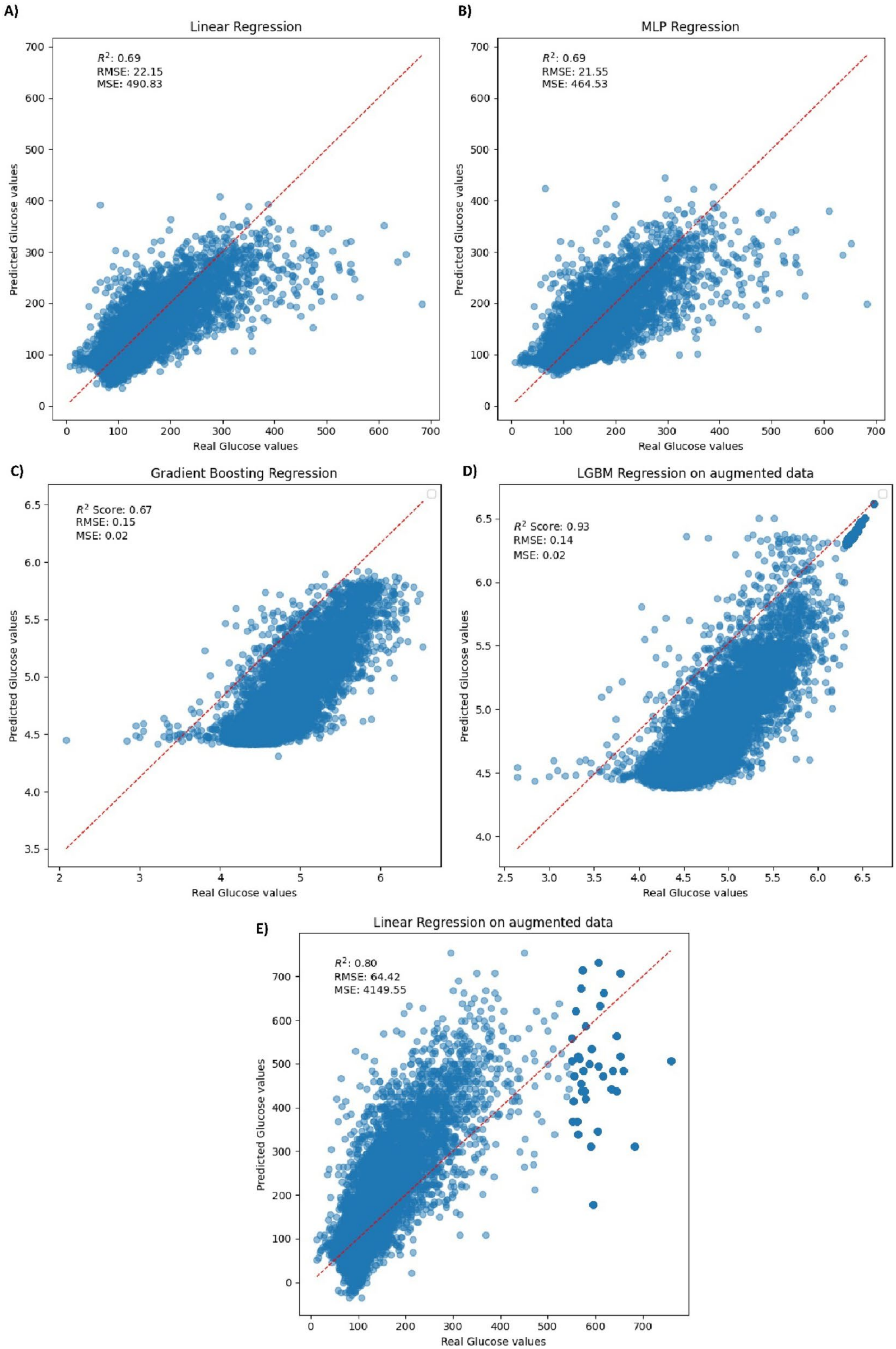


Fig. 6. (**A**) Linear Regression (raw, 197 k), (**B**) MLP Regression (raw, 197 k), (**C**) Gradient Boosting Regression (logarithmic transformed, 215 k), (**D**) LGBM Regression (oversampled,215 k), (**E**) Linear Regression (oversampled,215 k).

feature screening[21]. Elevated HbA_{1c} and glucose levels in older age groups further reflect a prolonged state of hyperglycemia, the prevalence of diabetes increased with age.

The study analyzed a large and heterogeneous cohort of 197,180 Turkish patients, with feature engineering creating key variables such as age, glucose levels, and HbA_{1c}. Using Python package, 42 models were tested, and the MLP Regressor consistently outperformed others across all raw data chunks, achieving an R^2 value of up to 0.71. Both MLP and traditional Linear Regression models exhibit an R^2 value of 0.69 for 197 k data, indicating a moderate level of predictive accuracy. However, the MLP Regressor demonstrates slightly better performance, as evidenced by lower RMSE and MSE values. While Linear Regression relies on a single-layer linear equation, the MLP Regressor's neural network-based approach increased its capability to handle the complexity of this large biological dataset[22].

Applying data transformation techniques, such as logarithmic scaling, was instrumental in addressing skewness and enhancing model performance. This approach aligns with the findings of Kovatchev et al., who proposed a logarithmic blood glucose measurement scaling to normalize data distributions and align clinical and numerical centers of the glucose scale. The symmetrization technique described in Kovatchev's work offers a framework for validating statistical tests on blood glucose data by normalization of the data[23]. Likewise, among the tested algorithms, logarithmic and square-root transformations produced the most efficient, effectively reducing skewness in our original dataset.

The logarithmic transformation, in particular, enabled the Gradient Boosting Regressor to achieve a notable R^2 of 0.67 alongside the lowest RMSE value. A high R^2 value signifies that the model accounts for a substantial portion of the variance in the data, while a RMSE indicates minimal prediction errors. Together, these metrics demonstrate that the model performs effectively by providing both accuracy and precision in its predictions[24]. These results highlight the efficacy of this transformation for handling right-skewed data commonly observed in blood glucose dataset.

To further improve model performance, addressing the imbalanced nature of the dataset was critical, particularly for extreme glucose values corresponding to hypoglycemia. These cases often represent underrepresented but clinically significant conditions in glucose prediction datasets, as most of the data points fall within the normal glucose range. To tackle this challenge, Gaussian KDE was applied to analyze the density of glucose values. KDE, as a non-parametric technique, provided a smooth and continuous estimate of the probability density function[25]. This analysis revealed a low-density region around glucose values of 550, highlighting the need to reinforce this region through targeted oversampling.

To address this low-density glucose region, data augmentation technique specifically on oversampling was explored to increase observations in this critical range. KDE proved instrumental in guiding these efforts, offering a framework to create a balanced dataset and improve the performance of a new ensemble classifier model, which ultimately yielded higher ML scores as demonstrated[26]. These findings highlight the critical importance of accurately modeling the hypoglycemic regions, as they play a pivotal role in ensuring the model's real-world clinical applicability. Through oversampling, the proportion of glucose values below 550 mg/dL was reduced from 99.8 to 91.4%, significantly improving predictive accuracy. The LGBM Regressor achieved an impressive R^2 score of 0.93 with the lowest RMSE on logarithmic-transformed blood glucose data. This underscores the value of advanced data preprocessing and model selection in enhancing predictive performance compared to traditional methods, such as linear regression, with significant implications for diabetes management in clinical applications[27]. While traditional Linear Regression provides a valid baseline, its performance is notably lower, especially in terms of error metrics, supporting the usage of gradient-based ensemble models for clinical prediction. This emphasizes the added value of ML based approaches, particularly when used with data augmentation and feature transformation.

A study demonstrated that deep transfer learning combined with data augmentation improved glucose level prediction accuracy for only classification task in Type 2 diabetes mellitus patients by overcoming limitations of small, imbalanced datasets within a clinically relevant 1-h horizon[28]. However, our study focuses on predicting blood glucose levels directly from HbA_{1c}, a parameter readily available in routine clinical practice, as opposed to requiring continuous glucose monitoring data, which is resource-intensive and less accessible in many healthcare settings. The methodology achieved a higher R^2 and lower RMSE than previously reported and introduces novel preprocessing techniques, including logarithmic transformations and KDE-driven oversampling, specifically tailored to HbA_{1c} derived datasets.

The data acquisition spanned a broad cross-section of hospital departments, providing a naturalistic snapshot of routine clinical populations. Although patients' diagnoses and current medications were not recorded, the inclusion of individuals from various clinical contexts helps mitigate selection bias. While all patients were of Turkish nationality the sample size and departmental diversity lend strength to the internal validity of the study population within the regional healthcare system.

In developing the predictive model, we focused on three simple input features: age, gender, and HbA_{1c} levels, which are commonly available in clinical settings. This choice was made to create a quick, deployable solution that avoids dependency on real-time data such as diet, physical activity, or stress, which can be hard to capture consistently. To show feasibility, we created a prototype web application where users enter just these three values to receive an estimated glucose level. This straightforward approach allows for easy integration into existing clinical workflows and is suitable for outpatient settings. While richer contextual data could enhance the model's accuracy, our results show that reasonable accuracy can still be achieved with minimal input, making this method practical for widespread use.

Nonetheless, certain limitations persist. The absence of comorbidity profiles, medication data, and ethnic heterogeneity constrains the external generalizability of the findings. Future work should aim to expand the study in more demographically diverse and internationally representative cohorts, including integration of clinical histories and medication data. Additionally, external validation across multi-ethnic and multi-center cohorts is essential. Doing so would help refine the model's performance and applicability across varied patient populations and clinical settings.

A critical consideration for translating our model into clinical practice is its deployability in real-world healthcare systems. While our approach emphasizes minimal input requirements in web base application, using only age, sex, and HbA_{1c} for glucose estimation, we acknowledge that successful clinical deployment involves more than predictive accuracy. Integration with continuous glucose monitoring (CGM) systems, electronic medical records (EMRs), and clinical decision support tools requires compatibility with existing data infrastructure[29]. Additionally, ethical and regulatory requirements such as GDPR and HIPAA compliance, secure data transmission, patient consent, and auditability must be addressed to protect patient privacy[30]. Although our current model was developed to be practical, we recognize the importance of exploring secure implementation frameworks, real-time data integration workflows, and institutional approvals. These aspects will be prioritized in future healthcare settings.

One promising direction for future work is the development of personalized predictive models that integrate data from CGM systems. Such multimodal integration could enhance prediction accuracy for dynamic risk assessments for individual patients. While our current model focuses on broad use with minimal inputs, future versions could include CGM data for more frequent feedback and adaptive learning, particularly in intensive care or those on insulin treatment.

Materials and methods

Data collection

This study utilized a dataset comprising demographic and clinical data collected from 501,482 Turkish patients whose blood samples were analyzed at the Biochemistry Department of Istanbul Haseki Educational and Training Hospital with Istanbul Provincial Health Directorate permission. All participants were of Turkish nationality. Blood samples were obtained from patients across various hospital departments, with no records available regarding their clinical diagnoses or current medications. This approach ensured a heterogeneous sample that reflects routine hospital patient demographics. The data included the following variables: patient ID, registration date, age, gender, insulin, creatinine, protein (spot urine), microalbumin (spot urine), glucose and HbA_{1c} levels offering a robust and diverse pool of data for analysis.

All methods were carried out in accordance with relevant guidelines and regulations. The study protocol was reviewed and approved by the Ethics Committee of Istanbul Haseki Educational and Training Hospital (Approval No:68–2024). Informed consent was obtained from all subjects and/or their legal guardians before data collection.

Data preprocessing

Data was filtered to remove rows with missing glucose or HbA_{1c} values, resulting in a clean subset of 197,180 entries. The age column was corrected to ensure accurate integer representation, and one-hot encoding was applied to the categorical variable "Gender" to create a binary representation of female patients[31].

Exploratory data analysis

Exploratory data analysis was conducted using Python's data visualization libraries Seaborn and Matplotlib to visualize the distributions of glucose and HbA_{1c} levels and to generate plots. Histograms with Kernel Density Estimates (KDE) were generated to assess the normality of distribution of variables in the dataset. Correlation analysis was performed to evaluate the relationships between age, glucose, insulin, protein (spot urine), microalbumin (spot urine), creatinine and HbA_{1c} levels.

Feature engineering

Features included in the ML models were age, HbA_{1c}, and gender. Raw and transformed glucose values (e.g., Logarithmic, Box-Cox, etc.) were used as the target variable across different model training sessions.

Machine learning model evaluation

For predictive modeling, the dataset was split into training and testing subsets using an 80:20 train-test split. The LazyPredict library facilitated the comparison of multiple regression models, including MLP Regressor, XGBoost, ElasticNetCV, and Lasso. Due to memory limitations encountered during analysis, we employed the SLURM workload manager to optimize resource allocation on a high-performance computing (HPC) cluster. The dataset was divided into four chunks, with each processed independently to assess model performance. The best model from these evaluations was then tested on the full dataset of 197,000 samples. Additionally, fivefold cross-validation was conducted to assess the model's robustness. RMSE was calculated for each fold, and the average RMSE was reported.

Data augmentation

To further enhance model performance, 6 different transformations were applied to the blood glucose values, including Logarithmic, Square-Root, Box-Cox, Reciprocal, Exponential, and Yeo-Johnson transformations (Table 4). Each transformed dataset was subjected to the same modeling process to evaluate the effectiveness of the transformations on predictive accuracy[32].

KDE based oversampling data

To identify outliers in the glucose data, we applied the Interquartile Range (IQR) method. The first quartile (Q1) and the third quartile (Q3) were calculated as follows [26]: Q1 represents the median of the lower half of the ordered dataset, while Q3 denotes the median of the upper half. Q2 serves as the overall median of the entire

Transformation	*Formula*	Main characteristics	Biological application
Logarithmic	$y' = \log(y)$	Suitable for large ranges	Gene expression analysis[33]
Square Root	$y' = \sqrt{y}$	Reduces right skewness	Cell growth rate analyses[34]
Box-Cox	$y\left(\lambda\right) = \frac{y^{\lambda}-1}{\lambda}, if \lambda \neq 0$	Normalizes data	Normalize metabolite levels for metabolomics[35]
Reciprocal	$y' = y^{-1}$	Effective compression of large values	Enzyme kinetics modeling[36]
Exponential	$y' = e^{y}$	Increase variance for low values	Cell proliferation[37]
Yeo-Johnson	$y\left(\lambda\right) = \frac{(y+1)^{\lambda}-1}{\lambda}, if \lambda >\neq 0$ $y\left(\lambda\right) = \log\left(y+1\right), if \lambda = 0$	Positive and negative data normalization	Biochemical signal analysis[38]

Table 4. Commonly used data transformations and their biological application.

dataset. The interquartile range (IQR) is calculated as the difference between Q3 and Q1. Outliers are identified as data points that fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$, indicating values that are significantly higher or lower than most of the data.

Following outlier removal, we employed Gaussian Kernel Density Estimation to model the distribution of glucose levels and computed density, using a bandwidth of 5, estimated across the glucose data. To identify regions of low density, we established a threshold based on the 25th percentile of the density estimates[39]. The formula for the Gaussian Kernel density estimator is as follows:

$$\hat{f}_h\left(x\right) = \frac{1}{nh\sigma}\frac{1}{\sqrt{2\pi}}\sum_{i=1}^{n}\exp\left(\frac{\left(-x-x_i\right)^2}{2h^2\sigma^2}\right)$$

where n denotes number of datapoints, σ is standard deviation, x_i is datapoint and x is function input variable.

To mitigate imbalances and improve model accuracy in critical hyperglycemic ranges (>550 mg/dL), we employed a KDE based random oversampling strategy. After determining the KDE based threshold, glucose > 550 mg/dL, we implemented random oversampling, whereby minority samples in the low-density region were duplicated with replacement to balance the dataset prior to model training. This process is illustrated in Fig. 7.

Statistical analysis

Model performance metrics, including R^2, Root Mean Square Error (RMSE), and Mean Square Error (MSE) were calculated to assess the accuracy of each model across different transformations and data chunks. All analysis done using Python 3.12 version. Model hyperparameter tuning was performed using GridSearchCV. Parameter ranges were defined for n_estimators, max_depth, learning_rate, and subsample. More information on was added: Model training was performed on a laptop with an Intel i7 CPU and 16 GB of RAM. No GPU was used.

Web application

The final ML predictive model was deployed on Heroku, a cloud-based platform, enabling the implementation of a fully cloud-based web application. A web-based application for glucose prediction is available on the Heroku server (https://glucosehba1c-02dd36d2b397.herokuapp.com/) (accessed on 22 January 2026). The code for reproducing the workflow and application can be also found at tmuhlise.github.io/app_glucose.py at main · tmuhlise/tmuhlise.github.io (accessed on 22 January 2026), with the specific application code available at tmuhlise/tmuhlise.github.io: Glucose prediction from HbA1c (accessed on 22 January 2026), ensuring long-term reproducibility beyond the current Heroku deployment.

Conclusion

Using a blood glucose test or a point-of-care testing (POCT) glucometer is a straightforward method for measuring blood glucose levels. However, ML models offer the unique advantage of integrating data from various sources, such as HbA_{1c}, diet, and urine protein, to provide personalized insights that traditional testing methods cannot achieve. In this study, a comprehensive comparative analysis revealed that the LGBM model was the most effective in predicting blood glucose levels. By employing a combination of logarithmic transformation, targeted oversampling, and the LGBM Regressor, our approach addresses challenges of data imbalance while ensuring high predictive precision.

Future work could build on these findings by exploring alternative oversampling techniques, integrating more clinical features (e.g., comorbidities, medication usage) to make the model scalable for larger, more diverse populations. Expanding the dataset to include multiethnic cohorts and international clinical settings would be particularly valuable for improving the model's external validity. Additionally, testing the model in real-world clinical environments through EHR integration would further support its deployment potential. Such advancements could significantly enhance the model's robustness, generalizability, and clinical utility in predicting glucose levels across a wide range of patient populations.

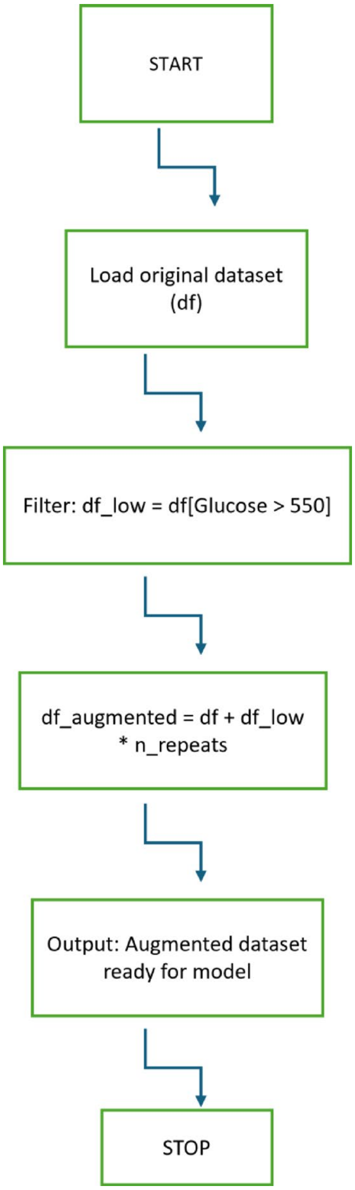


Fig. 7. Oversampling strategy diagram for mitigating imbalances in hyperglycemic ranges.

Data availability

The clinical data used in this study were obtained from Haseki Training and Research Hospital and are not publicly available due to institutional privacy policies and ethical considerations. Access to the data may be granted upon reasonable request and with permission from Haseki Hospital and relevant ethical committees. The source code used for model development and web application deployment is openly available on GitHub at: https://github.com/tmuhlise/tmuhlise.github.io (accessed on 22 January 2026). The web-based application can be accessed at: https://glucosehba1c-02dd36d2b397.herokuapp.com/ (accessed on 22 January 2026). All resources provided ensure long-term accessibility and reproducibility of the computational workflow beyond the current Heroku deployment.

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Author contributions

T.M.O. analyzed the data, formulated the code, wrote the paper, I.Y. edited the paper, M.K supervised and coordinated the research project.

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Declarations

Competing interests
The authors declare no competing interests.

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Consent to participate
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